

Reproduce-then-Extend — State Policy Liberalism (American State Politics) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Caughey, D. & Warshaw, C. (2016). “The Dynamics of State Policy Liberalism, 1936–2014.” *American Journal of Political Science* 60(4):899–913. Building on the state-representation tradition (Erikson, Wright & McIver, *Statehouse Democracy*, 1993), the paper measures each state’s **policy liberalism** and links it to public opinion and state political conditions – a **cross-sectional OLS** of policy liberalism on state characteristics. This example illustrates **regularization and variable selection** in a small-sample ($n = 50$) design, where predictors are many relative to observations.

1 Background

A central question in American state politics is why some states adopt systematically more **liberal policies** than others. The state-representation literature argues that policy tracks **public opinion** (the “opinion–policy link”), mediated by **party control** and the organized strength of liberal and conservative constituencies. Caughey & Warshaw construct a dynamic measure of state policy liberalism and show it responds to mass preferences. Here we treat the **cross-section of the 50 states** as the unit of analysis and ask which state characteristics predict policy liberalism – the setting the state-politics literature has studied with ordinary regression for decades.

2 Data and codebook

Unit of analysis: a U.S. **state** (cross-section averaged over 2005–2012); $n = 50$. Data are drawn from the **Correlates of State Policy Project** (Jordan & Grossmann 2020), which compiles the standard state-politics measures.

Variable	Definition
pollib_median	state policy liberalism (Caughey–Warshaw), higher = more liberal (outcome)
mood	public policy mood (higher = more liberal preferences)
citi6013	citizen ideology (Berry et al.), higher = more liberal
ranney4_control	Ranney index of Democratic party control (higher = more Democratic)
union_density	percent of workers unionized
evangelical_pop	evangelical share of the population
gini_coef	income inequality (Gini coefficient)
gsppcap	gross state product per capita

3 Descriptive analysis

We first load the data and get to know it thoroughly before any modeling. The data-loading step is unchanged from the reproduction below; the descriptive analysis operates on the loaded data frame `d`.

```
d <-  
  ↪ read.csv("https://raw.githubusercontent.com/bdesmarais/intro-ml-statistical-horizons-4day/main/tuto  
preds <- setdiff(names(d), c("st", "pollib_median"))  
cat(sprintf("%d states x %d predictors (p/n = %.2f)\n",  
            nrow(d), length(preds), length(preds) / nrow(d)))
```

```
## 50 states x 7 predictors (p/n = 0.14)
```

3.1 Dataset and provenance

Unit of analysis. One row per U.S. **state**, a pure cross-section averaged over roughly 2005–2012, so $n = 50$. There is no time dimension and no sampling: this is the *entire population* of states, which is exactly why the small- n problem is structural rather than fixable by collecting more data. With 7 predictors and 50 states, the predictor-to-observation ratio is 0.14 – modest in absolute terms, but large enough that ordinary least squares has few states per coefficient and begins to over-fit. That tension is the entire point of this Day-3 example.

Source. Variables are drawn from the **Correlates of State Policy Project** (Jordan & Grossmann 2020), which compiles the standard state-politics measures; the outcome is the Caughey–Warshaw (2016) dynamic policy-liberalism index.

The variable dictionary below lists the outcome and all 7 predictors, their role, type, and coding direction.

```
dict <- data.frame(  
  Variable = c("pollib_median", "mood", "citi6013", "ranney4_control",  
              "union_density", "evangelical_pop", "gini_coef", "gsppcap"),  
  Role = c("outcome", rep("predictor", 7)),  
  Type = "numeric",  
  Description = c(  
    "Policy liberalism (Caughey-Warshaw); higher = more liberal",  
    "Public policy mood; higher = more liberal preferences",  
    "Citizen ideology (Berry et al.); higher = more liberal",  
    "Ranney index of Democratic party control; higher = more Dem.",  
    "Percent of workers unionized",  
    "Evangelical share of the population",  
    "Income inequality (Gini coefficient)",  
    "Gross state product per capita (US$)")  
kable(dict)
```

Variable	Role	Type	Description
pollib_median	outcome	numeric	Policy liberalism (Caughey-Warshaw); higher = more liberal
mood	predictor	numeric	Public policy mood; higher = more liberal preferences
citi6013	predictor	numeric	Citizen ideology (Berry et al.); higher = more liberal
ranney4_control	predictor	numeric	Ranney index of Democratic party control; higher = more Dem.
union_density	predictor	numeric	Percent of workers unionized
evangelical_pop	predictor	numeric	Evangelical share of the population
gini_coef	predictor	numeric	Income inequality (Gini coefficient)
gsppcap	predictor	numeric	Gross state product per capita (US\$)

All eight variables are continuous. There are no categorical or binary predictors, which keeps the descriptive

work focused on distributions and correlations.

3.2 The outcome in depth

`pollib_median` is a standardized latent index, so it is centered near zero and takes both signs (negative = more conservative than average, positive = more liberal). We examine its full distribution and shape.

```
p1 <- ggplot(d, aes(pollib_median)) +
  geom_histogram(aes(y = after_stat(density)), bins = 12,
    fill = "#a6cee3", colour = "white") +
  geom_density(colour = "#1f78b4", linewidth = 0.7) +
  labs(x = "policy liberalism", y = "density", title = "Outcome distribution")
p2 <- ggplot(d, aes(y = pollib_median)) +
  geom_boxplot(fill = "#a6cee3", width = 0.4) +
  labs(y = "policy liberalism", x = NULL, title = "Boxplot")
p1 + p2
```

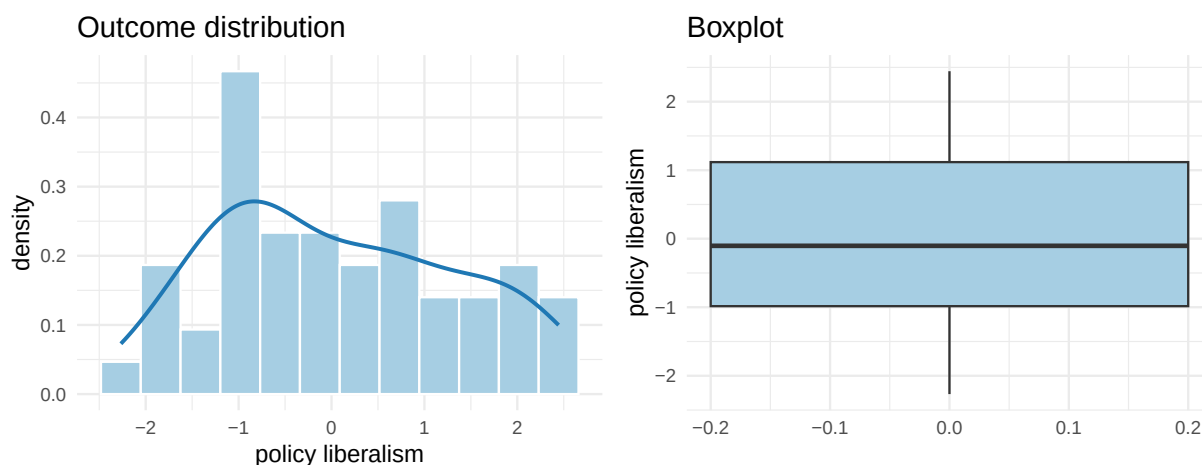


Figure 1: Distribution of state policy liberalism: histogram with density overlay (left) and boxplot (right).

```
stat_row <- function(x) {
  q <- quantile(x, c(.25, .5, .75))
  c(n = length(x), mean = mean(x), sd = sd(x), min = min(x),
    Q1 = q[1], median = q[2], Q3 = q[3], max = max(x),
    IQR = q[3] - q[1], skew = skewness(x), kurt = kurtosis(x))
}
out_tab <- t(stat_row(d$pollib_median))
rownames(out_tab) <- "pollib_median"
kable(round(out_tab, 2))
```

	n	mean	sd	min	Q1.25%	median.50%	Q3.75%	max	IQR.75%	skew	kurt
pollib_median	50	0.06	1.29	-2.27	-0.98	-0.1	1.12	2.44	2.1	0.23	-1.12

Shape. The outcome is close to **symmetric** (skewness 0.23) and mildly **platykurtic** (excess kurtosis -1.12) – flatter than a normal, with mass spread fairly evenly from conservative to liberal states rather than bunched at a mode. There is no zero-inflation, no floor or ceiling, and no obvious outlier: the boxplot whiskers cover the full range with no points beyond them. A response transform is **not warranted**; we model `pollib_median` on its natural scale, as the published OLS does.

3.3 Univariate predictor distributions

With only seven predictors we can profile each one. The faceted histograms show the marginal shape of every predictor, and the summary table quantifies it.

```
d |>
  select(all_of(preds)) |>
  pivot_longer(everything(), names_to = "var", values_to = "val") |>
  ggplot(aes(val)) +
  geom_histogram(bins = 12, fill = "#b2df8a", colour = "white") +
  facet_wrap(~ var, scales = "free", ncol = 4) +
  labs(x = NULL, y = "count")
```

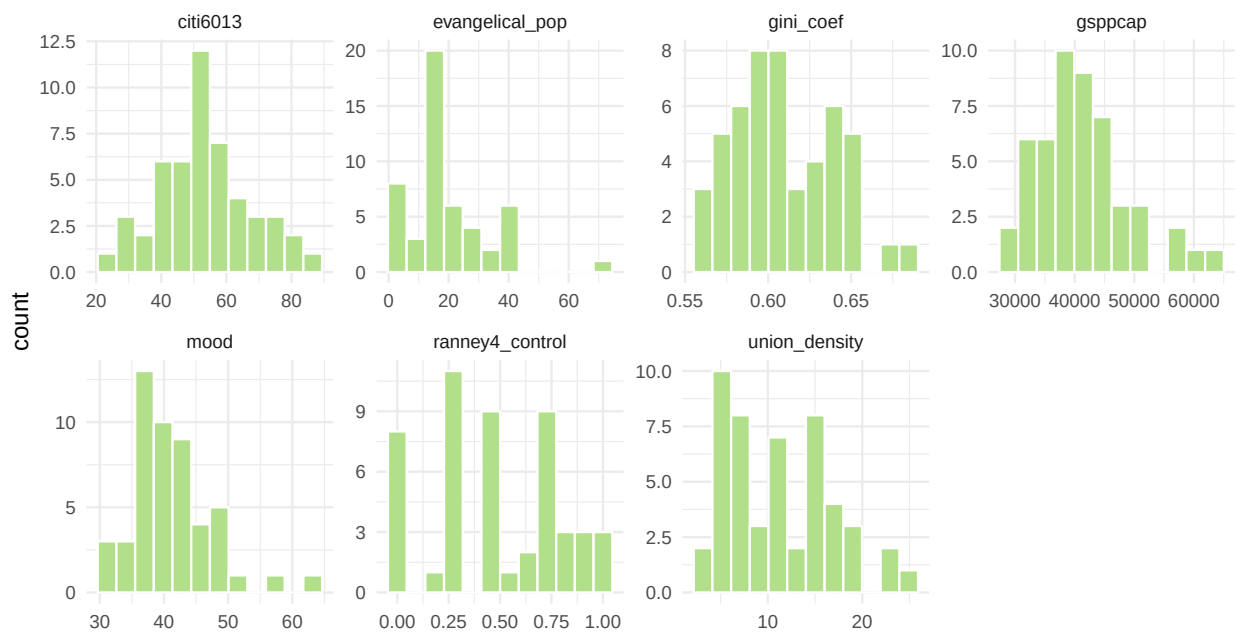


Figure 2: Marginal distributions of the seven predictors.

```
pred_tab <- t(sapply(d[preds], function(x) {
  q <- quantile(x, c(.25, .5, .75))
  c(n = length(x), n_miss = sum(is.na(x)), mean = mean(x), sd = sd(x),
    min = min(x), Q1 = q[1], median = q[2], Q3 = q[3], max = max(x),
    skew = skewness(x))
}))
colnames(pred_tab) <- c("n", "n_miss", "mean", "sd", "min",
  "Q1", "median", "Q3", "max", "skew")
kable(round(pred_tab, 2))
```

	n	n_miss	mean	sd	min	Q1	median	Q3	max	skew
citi6013	50	0	53.36	14.70	23.34	43.52	52.52	61.61	86.52	0.21
ranney4_control	50	0	0.49	0.31	0.00	0.31	0.50	0.75	1.00	-0.09
evangelical_pop	50	0	19.55	13.24	2.90	12.93	15.95	25.90	71.40	1.44
union_density	50	0	11.23	5.60	3.14	6.13	10.90	15.07	24.91	0.54
gini_coef	50	0	0.61	0.03	0.56	0.59	0.60	0.63	0.68	0.41
mood	50	0	41.23	5.96	31.14	37.37	40.65	43.62	63.17	1.21
gsppcap	50	0	41596.20	7747.83	28955.83	36696.75	40911.67	45499.71	63450.17	0.81

The public-opinion and party-control measures (`mood`, `citi6013`, `ranney4_control`) are roughly symmetric. The socio-economic predictors are **right-skewed**: `evangelical_pop` (skew 1.44), `mood`, and `gsppcap` all have a long upper tail driven by a handful of extreme states (high-evangelical Southern states; high-income outliers). The predictor scales differ by orders of magnitude (`gsppcap` in the tens of thousands vs. proportions near zero), which is exactly why **glmnet standardizes predictors internally** before penalizing – otherwise the penalty would fall almost entirely on the large-scale variables.

3.4 Missing data

```
miss <- data.frame(
  variable = c("pollib_median", preds),
  n_missing = sapply(d[c("pollib_median", preds)], function(x) sum(is.na(x)))
miss$pct_missing <- round(100 * miss$n_missing / nrow(d), 1)
kable(miss, row.names = FALSE)
```

variable	n_missing	pct_missing
pollib_median	0	0
citi6013	0	0
ranney4_control	0	0
evangelical_pop	0	0
union_density	0	0
gini_coef	0	0
mood	0	0
gsppcap	0	0

```
cat(sprintf("Complete cases: %d of %d states (%.0f%%).\n",
  sum(complete.cases(d)), nrow(d),
  100 * mean(complete.cases(d))))
```

```
## Complete cases: 50 of 50 states (100%).
```

The data are **fully complete**: every state has every variable, so there is no co-missingness pattern to display and no listwise deletion in the reproduction. This is expected for an administrative population-level compilation – the Correlates of State Policy Project reports each measure for all 50 states – and means the missing-data mechanism is a non-issue here. All 50 states enter every model.

3.5 Bivariate relationships with the outcome

We next look at how each predictor co-moves with policy liberalism. Because every predictor is numeric, we use scatterplots with a linear smoother and annotate each panel with its Pearson correlation `r`.

```
long <- d |>
  select(pollib_median, all_of(preds)) |>
  pivot_longer(all_of(preds), names_to = "var", values_to = "val")
rlab <- long |>
  group_by(var) |>
  summarise(r = cor(val, pollib_median), .groups = "drop") |>
  mutate(lab = sprintf("r = %.2f", r))
ggplot(long, aes(val, pollib_median)) +
  geom_point(colour = "#1f78b4", alpha = 0.6, size = 1) +
  geom_smooth(method = "lm", se = FALSE, colour = "#e31a1c",
    linewidth = 0.6) +
  geom_text(data = rlab, aes(x = -Inf, y = Inf, label = lab),
```

```
hjust = -0.1, vjust = 1.4, size = 2.6, inherit.aes = FALSE) +
facet_wrap(~ var, scales = "free_x", ncol = 4) +
labs(x = NULL, y = "policy liberalism")
```

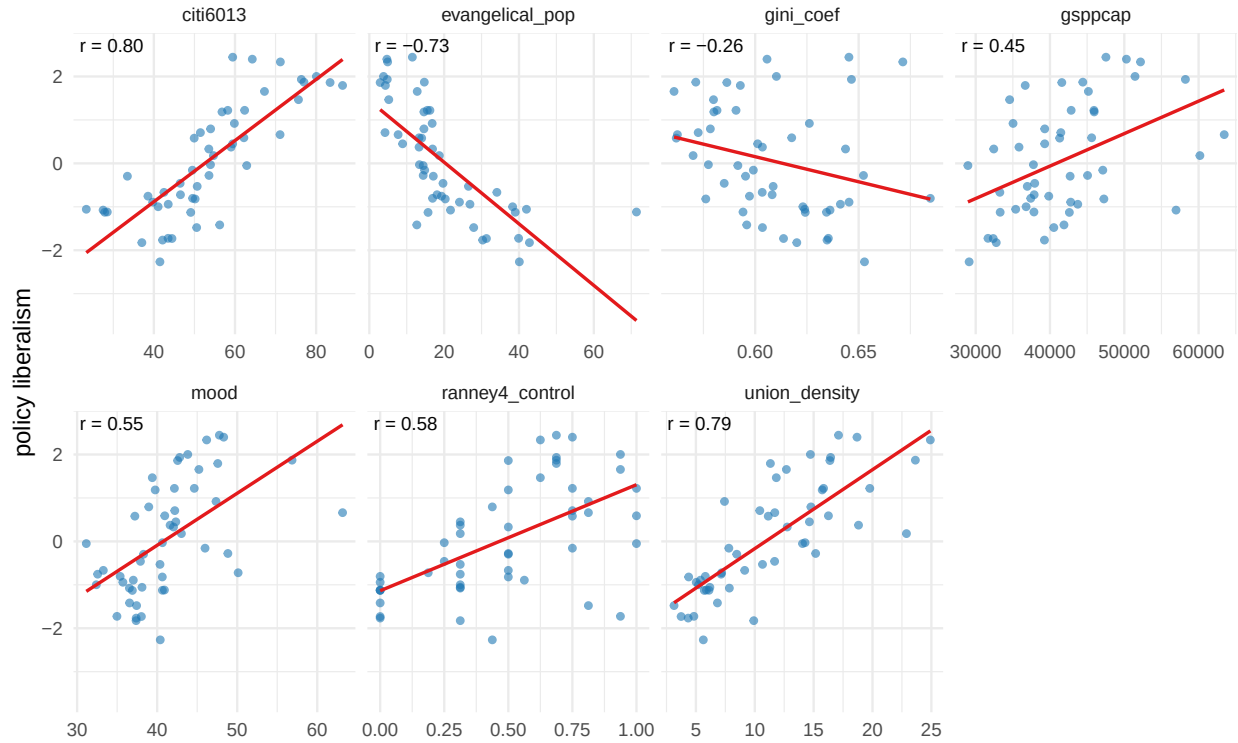


Figure 3: Policy liberalism vs each predictor, with linear fit and Pearson r.

```
marg <- long |>
  group_by(var) |>
  summarise(r = cor(val, pollib_median),
            r2 = r^2, .groups = "drop") |>
  arrange(desc(abs(r)))
kable(marg, digits = 2,
      col.names = c("predictor", "r with outcome", "r-squared"))
```

predictor	r with outcome	r-squared
citi6013	0.80	0.64
union_density	0.79	0.62
evangelical_pop	-0.73	0.53
ranney4_control	0.58	0.34
mood	0.55	0.31
gsppcap	0.45	0.20
gini_coef	-0.26	0.07

Reading the associations. The opinion-policy link dominates: **public mood** and **citizen ideology** are the strongest marginal predictors of policy liberalism (large positive r), followed by **Democratic party control** and **union density**. **Evangelical population** is the strongest *negative* correlate – more evangelical states are more conservative. Income inequality and per-capita product are weaker, more ambiguous marginal predictors. These raw associations preview the signs the reproduction recovers.

3.6 Multivariate structure and collinearity

Marginal correlations can mislead when predictors are themselves correlated. We examine the full predictor correlation matrix and the variance-inflation factors (VIFs) from the reproduction model.

```
cm <- cor(d[preds])
cm_long <- as.data.frame(as.table(cm))
names(cm_long) <- c("v1", "v2", "r")
ggplot(cm_long, aes(v1, v2, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), size = 2.4) +
  scale_fill_gradient2(low = "#e31a1c", mid = "white",
    high = "#1f78b4", midpoint = 0, limits = c(-1, 1)) +
  labs(x = NULL, y = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Figure 4: Correlation heatmap of the seven predictors.

```
# VIFs via auxiliary regressions (car is not installed)
vif <- sapply(preds, function(p) {
  aux <- lm(as.formula(paste(p, "~ .")), data = d[preds])
  1 / (1 - summary(aux)$r.squared)
})
kable(data.frame(predictor = preds, VIF = round(vif, 2)),
  row.names = FALSE)
```

predictor	VIF
citi6013	3.56
ranney4_control	1.45
evangelical_pop	2.78
union_density	1.91
gini_coef	1.11
mood	1.71

predictor	VIF
gsppcap	1.60

Only one predictor pair exceeds the $|r| > 0.7$ flag: **citizen ideology and evangelical population** correlate about **-0.76** (conservative, high-evangelical states have low citizen-ideology scores). No VIF is alarmingly high – the largest is `citi6013` at 3.6, well under the usual 10 rule of thumb – so the collinearity is moderate, not pathological. But combined with only 50 states, even moderate collinearity inflates OLS coefficient variance, which is precisely the instability that **regularization** is built to tame in the sections that follow.

3.7 From description to modeling

The descriptive picture sets up the rest of the tutorial cleanly. The outcome is symmetric and well-behaved (no transform needed); the predictors carry a clear opinion-policy signal but differ wildly in scale (so `glmnet` standardizes them); the data are complete (all 50 states model-able); and with seven correlated predictors over just 50 states, OLS has little slack. We now reproduce the published regression, then show that penalized and best-subset methods predict new states more accurately while naming the compact set of characteristics that drive policy liberalism.

4 Reproduce the published regression

The state-representation model regresses policy liberalism on public opinion and state political conditions.

```
fit <- lm(pollib_median ~ ., data = d[, c("pollib_median", preds)])
round(summary(fit)$coefficients, 3)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-3.195	2.180	-1.466	0.150
## citi6013	0.024	0.011	2.157	0.037
## ranney4_control	0.670	0.338	1.980	0.054
## evangelical_pop	-0.018	0.011	-1.642	0.108
## union_density	0.090	0.021	4.235	0.000
## gini_coef	0.001	3.072	0.000	1.000
## mood	0.019	0.019	0.991	0.327
## gsppcap	0.000	0.000	0.357	0.723

Confirmation against the literature. In-sample $R^2 = 0.81$, and the **opinion-policy link** is recovered: more liberal **public mood** and **citizen ideology**, stronger **Democratic control**, and higher **union density** all predict more liberal policy, while a larger **evangelical** population predicts more conservative policy – the signs the state-politics literature reports.

5 Regularization & variable selection

With only **50 states** and **7 predictors**, ordinary least squares has few observations per coefficient and **over-fits**: it fits the 50 states well but predicts new states poorly. Penalized regression – lasso (L_1), ridge (L_2), elastic net – shrinks the coefficients to trade a little bias for much lower variance.

```
X <- as.matrix(d[, preds]); y <- d$pollib_median
set.seed(2026)
lasso <- cv.glmnet(X, y, alpha = 1); ridge <- cv.glmnet(X, y, alpha = 0); enet <-
  ↪ cv.glmnet(X, y, alpha = 0.5)
n_keep <- sum(coef(lasso, s = "lambda.1se")[-1] != 0)
cat(sprintf("At lambda.1se the lasso keeps %d of %d predictors (the rest are set to
  ↪ exactly 0).\n",
```



```
n_keep, length(preds)))
```

At lambda.1se the lasso keeps 4 of 7 predictors (the rest are set to exactly 0).

```
data.frame(
  OLS      = round(coef(fit)[-1], 3),
  Lasso     = round(as.numeric(coef(lasso, s = "lambda.1se"))[-1], 3),
  Ridge     = round(as.numeric(coef(ridge, s = "lambda.1se"))[-1], 3),
  ElasticNet = round(as.numeric(coef(enet, s = "lambda.1se"))[-1], 3),
  row.names = preds)
```

```
##           OLS  Lasso  Ridge ElasticNet
## citi6013    0.024  0.025  0.016      0.020
## ranney4_control 0.670  0.046  0.525      0.283
## evangelical_pop -0.018 -0.007 -0.015     -0.013
## union_density  0.090  0.068  0.047      0.063
## gini_coef      0.001  0.000 -1.461      0.000
## mood          0.019  0.000  0.022      0.005
## gsppcap       0.000  0.000  0.000      0.000
```

Which fits best out of sample? With $n = 50$ a single split is noisy, so we **repeat a 70/30 split 50 times**: each replicate tunes the penalty by cross-validation on the training states and scores once on the held-out states. Penalized models are scored at `lambda.min` (the loss-minimizing penalty).

```
rmse <- function(p, yt) sqrt(mean((yt - p)^2))
REPS <- 50; alphas <- c(Lasso = 1, Ridge = 0, ElasticNet = 0.5)
M <- matrix(NA, REPS, 4, dimnames = list(NULL, c("OLS", names(alphas))))
Xdf <- as.data.frame(X)
for (r in 1:REPS) {
  set.seed(2025 + r)
  te <- sample(nrow(X), round(0.30 * nrow(X))); tr <- setdiff(seq_len(nrow(X)), te)
  ols <- lm(y[tr] ~ ., data = Xdf[tr, ]); M[r, "OLS"] <- rmse(predict(ols, Xdf[te, ]),
  ↪ y[te])
  for (a in names(alphas)) {
    cv <- cv.glmnet(X[tr, ], y[tr], alpha = alphas[a])
    M[r, a] <- rmse(as.numeric(predict(cv, X[te, ], s = "lambda.min")), y[te])
  }
}
round(colMeans(M), 3)                                     # mean held-out RMSE over 50
↪ splits
```

```
##           OLS      Lasso      Ridge ElasticNet
##           0.736      0.721      0.688      0.704
```

```
best <- names(which.min(colMeans(M)[names(alphas)]))
cat(sprintf("\nBest penalization: %s. Held-out RMSE: %s = %.3f vs OLS %.3f (%.0f%%
  ↪ lower).\n",
            best, best, min(colMeans(M)[names(alphas)]), colMeans(M)[ "OLS" ],
            100 * (colMeans(M)[ "OLS" ] - min(colMeans(M)[names(alphas)])) /
            ↪ colMeans(M)[ "OLS" ]))
```

```
##
## Best penalization: Ridge. Held-out RMSE: Ridge = 0.688 vs OLS 0.736 (6% lower).
```

Because there are so few states per predictor, unpenalized OLS over-fits and the penalized models **predict new states more accurately**, while the lasso simultaneously **selects** the handful of state characteristics that carry the signal.

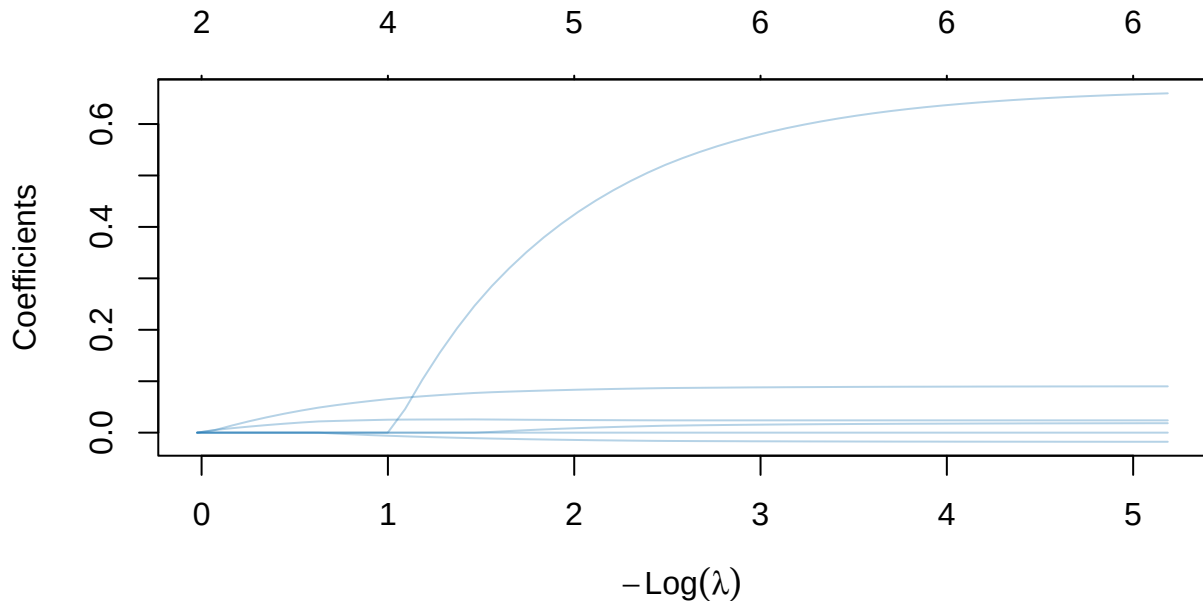


Figure 5: Lasso coefficient paths vs the penalty (log lambda); paths reaching zero are de-selected. The dashed line marks the CV-optimal penalty.

6 Best-subset selection with ABESS

Lasso reaches a sparse model through shrinkage. **Best-subset selection** instead searches directly for the subset of predictors that fits best; the **adaptive best-subset (ABESS)** algorithm does so efficiently and chooses the subset size automatically.

```
library(abess)
ab <- abess(X, y)
cat(sprintf("ABESS selects a best subset of %d of %d predictors:\n", ab$best.size,
  ↪ length(preds)))
```

ABESS selects a best subset of 4 of 7 predictors:

```
cf <- as.numeric(coef(ab, support.size = ab$best.size))[-1]
data.frame(predictor = preds, coef = round(cf, 3))[order(-abs(cf)), ]
```

```
##      predictor    coef
## 2 ranney4_control 0.691
## 4 union_density  0.097
## 1      citi6013  0.026
## 3 evangelical_pop -0.019
## 5      gini_coef  0.000
## 6          mood  0.000
## 7      gsppcap   0.000
```

Best subset and the lasso agree on the core drivers of state policy liberalism – public opinion, party control, and organized labor – reached by direct search rather than shrinkage.

7 Takeaway

This example illustrates **regularization in a small-sample design**. State-politics regressions have few observations (50 states) relative to their predictors, so ordinary regression over-fits; lasso/elastic-net

(`glmnet`) and best-subset selection (`abess`) predict new states more accurately **and** name the compact set of characteristics that drive policy liberalism.

8 Recommended exercises

1. Add a couple more state characteristics from the Correlates of State Policy data and see whether the held-out gain over OLS grows as the predictor-to-state ratio rises.
2. Compare `lambda.min` and `lambda.1se` predictions: how much accuracy does the sparser 1-SE model give up?
3. Refit predicting a *specific* policy (e.g., minimum wage) instead of the liberalism index and report which predictors the lasso keeps.